Movie Ticketing System

Software Requirements Specification

<Version 1.0>

<02/19/26>

Group #19
Michelle Khaliqi, Nandana Manu,
Katarina Cedolin

Prepared for
CS 250- Introduction to Software Systems
Instructor: Dr. Gus Hanna
Spring 2026

Revision History

Date	Description	Author	Comments
2/15	Version 1.0	Group 19	< SRS 1.0 >
2/19	Update Version 1.0	Group 19	< SRS 1.0 >

Document Approval

The following Software Requirements Specification has been accepted and approved by the following:

Signature	Printed Name	Title	Date
	<Your Name>	Software Eng.	
	Dr. Gus Hanna	Instructor, CS 250	

Table of Contents

Revision History ii

Document Approval ii

1. Introduction 1

1.1 Purpose 1
1.2 Scope 1
1.3 Definitions, Acronyms, and Abbreviations 1
1.4 References 2
1.5 Overview 2

2. General Description 2

2.1 Product Perspective 2
2.2 Product Functions 3
2.3 User Characteristics 3
2.4 General Constraints 4
2.5 Assumptions and Dependencies 4

3. Specific Requirements 4

3.1 External Interface Requirements 5
3.1.1 User Interfaces 5
3.1.2 Hardware Interfaces 5
3.1.3 Software Interfaces 6
3.1.4 Communications Interfaces 6
3.2 Functional Requirements 6
3.2.1 <Concurrent User Support> 6
3.2.2 <Web Browser Access> 7
3.2.3 <Bot Detection and Protection> 7
3.2.4 <Database Interface Integration> 8
3.2.5 <Ticket Purchase Constraints> 8
3.2.6 <Administrator Mode Support> 9
3.2.7 <Customer Feedback System> 10
3.2.8 <Discount Ticket Support> 10
3.2.9 <Online Review Scraping + Integration> 10
3.2.10 <Multilingual Support> 11
3.2.11 <Loyalty Account Management> 11
3.3 Use Cases 12
3.3.1 Use Case #1 12
3.3.2 Use Case #2 12
3.3.3 Use Case #3 13
3.4 Classes / Objects 14
3.4.1 <User> 14
3.4.2 <Ticket> 14
3.4.3 <Movie> 15
3.4.4 <Administrator> 15
3.4.5 <Payment> 16
3.4.6 <Feedback> 16
3.4.7 <Theatre> 16
3.5 Non-Functional Requirements 17
3.5.1 Performance 17
3.5.2 Reliability 17
3.5.3 Availability 17
3.5.4 Security 17
3.5.5 Maintainability 17
3.5.6 Portability 18

1. Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) document is to provide a clear outline of the requirements for building a Movie Ticketing System, including the system's functionality and constraints. This document is intended for the developers responsible for creating the Movie Ticketing System.

1.2 Scope

The Movie Ticketing System is to be accessed by customers and primary users of the system, which include system administrators, cinema staff, and movie theater customers.

The Movie Ticketing System is a software application designed for customers to browse through available movies, select designated showtimes, reserve seats, and purchase available tickets online. After confirmation and payment, customers will receive an email that includes their receipt and a personal QR code that is ready to be scanned by cinema staff in person. Additionally, the system will allow administrators to update movie listings, showtimes, and pricing information. Cinema staff will confirm bookings, view prices, and monitor showtimes without access to modify data. The Movie Ticketing System will not stream movies or provide services for snacks.

The goal of this Movie Ticketing System is to be a faster and more convenient method of purchasing movie tickets, preventing long lines of customers who purchase tickets in person. The system aims to handle high user traffic hours of purchases, update seat availability so the same seat is not purchased by two individuals, and securely process online payments. The constraints for this section includes our deadline of May 7, 2026 for the document.

1.3 Definitions, Acronyms, and Abbreviations

The following terms, acronyms, and abbreviations are used in this SRS and are defined here to support consistent interpretation of the requirements.

API: Application Programming Interface
CAPTCHA: Completely Automated Public Turing test to tell Computers and Humans Apart
HTTP: Hypertext Transfer Protocol
HTTPS: Hypertext Transfer Protocol Secure
HTTP(S): HTTP or HTTPS
IMDB: Internet Movie Database
MTBF: Mean Time Between Failures
QR: Quick Response
SRS: Software Requirements Specification

1.4 References

The following sources were used in preparation of the SRS:

Cornell University. (2019, Spring). *Use cases* [Lecture slides]. Cornell University, Department of Computer Science. https://www.cs.cornell.edu/courses/cs5150/2019sp/slides/7-use-cases.pdf

GeeksforGeeks. (2026). *Unified Modeling Language (UML) class diagrams*. GeeksforGeeks. https://www.geeksforgeeks.org/system-design/unified-modeling-language-uml-class-diagrams/

KrazyTech. (2025). *Sample software requirements specification (SRS) report – Airline database*. KrazyTech. https://krazytech.com/projects/sample-software-requirements-specificationsrs-report-airline-database

McMillan, J. *Requirements handout*. Clemson University, Department of Computing. https://people.computing.clemson.edu/~johnmc/courses/cpsc371/RequirementsHandout.pdf

Relevant Software. (2023). *Software requirements specification (SRS) document*. Relevant Software Blog. https://relevant.software/blog/software-requirements-specification-srs-document/

1.5 Overview

This Software Requirements Specification document outlines all the features and considerations necessary in developing the Movie Ticketing System. Section 2 provides a general description of the system, including the main functions, user characteristics, constraints, and assumptions. Section 3 lists the specific requirements, such as how users interact with the system, specific features of the system, and use cases. This SRS is organized to help the software developers understand the system and use the requirements listed to design and implement the software.

2. General Description

2.1 Product Perspective

The Movie Theater Ticketing System is a web-based software product that supports the end-to-end process of browsing movies, selecting showtimes, reserving tickets, purchasing tickets, and verifying tickets in person. The system complements in-person ticket sales, box office sales, and online purchases by ensuring all purchases reflect the same seat/ticket inventory in real time to prevent conflicts and double-booking. The initial deployment targets 20 theaters in

the San Diego area operating in the Pacific Time zone with a shared, centrally managed database partitioned by theater.

The system interacts with several external elements; customers, through a web interface, cinema staff, through ticket validation, administrators, through management functions for listings, schedules and pricing, and supporting services such as payment processing and email delivery for receipts and tickets. The theater network includes regular-auditorium theaters (general admission) and deluxe-auditorium theaters (reserved/assigned seating); seat inventory rules differ by auditorium type.

2.2 Product Functions

The proposed system will provide the following major functionalities:

- Movie discovery and browsing: Display to customer current and upcoming movies, descriptions, ratings, and showtimes by theater location.
- Showtime and seat selection: Allow users to choose a theater, date/time, and available seats from a seating map.
- Ticket purchase: Allow users to purchase tickets of several categories (Child, Adult, Senior, Discount if applicable) and reserve these seats with a checkout flow.
- Transaction confirmation: Provide a confirmation page and send a digital (email or text) receipt containing purchase details and a unique ticket number/barcode.
- Ticket validation: Allow cinema staff to confirm ticket validity and view showtime and seat details without editing listings.
- Administration and maintenance: Allow administrators to manage movies, showtimes, seat inventory, and pricing.
- Possible data collection for activity logs: Track and summarize sale counts and activity logs for administrative oversight.

2.3 User Characteristics

The system is expected to be used by three primary user groups, whose characteristics influence usability, accessibility, and interface design.

- Customers (general public): Vary widely in technical abilities. Will likely require a simple, mobile-friendly web interface with clear navigation, minimal steps to purchase, and understandable error messages. First-time users should not require instructions to carry out their intended actions. Large numbers of users at a time should be expected. Customers may purchase from mobile or desktop browsers; flows should remain simple under high concurrency.
- Cinema Staff: Frequent users during operating hours who prioritize speed and accuracy when verifying tickets and resolving entry issues. Training time is limited so workflows should be efficient and minimize common errors. Staff also handle exceptions such as in-person refunds and entry issues, so validation workflows should surface clear ticket status and show details quickly.
- System Administrators: Technically comfortable users who manage listings and operational data and are very familiar with the non-technical operations of the movie theater. They need dependable configuration tools, clear visibility into changes, and the

ability to quickly correct items such as showtime schedules and pricing. Admins may require override tools to correct customer purchase mistakes and adjust showtimes/theaters with minimal disruption.

2.4 General Constraints

The following constraints may limit design and implementation options.

- Browser-only access: The system is intended to run in modern web browsers and should not require users to install an application.
- Security and privacy considerations: The system processes sensitive user information and must incorporate the appropriate protections.
- Operational environment: The system must function during peak demand and be able to accommodate growing crowds. It must continue to maintain accurate seat availability to reduce double-booking.
- External services: Payment processing and email delivery will depend on third-party services.
- Schedule constraints: Delivery of the project milestones are bounded by course deadlines. (4 months)
- Budget constraints: Budget is limited to $500,000.

2.5 Assumptions and Dependencies

The SRS assumes the following conditions will hold; changes to these assumptions may require updating system requirements.

- Reliable internet access is available for customers and theater staff.
- Email delivery service is available and configured to send confirmations to customers after purchases.
- Payment processing service is available for online transactions.
- A centralized database is available to store movies, showtimes, seat inventory, users, transactions, and administrative data.
- Theater operational data is appropriately maintained by administrators.
- Ticket verification capabilities exists for staff, eg. a scanner or device capable of validating bookings.

3. Specific Requirements

3.1 External Interface Requirements

3.1.1 User Interfaces

- The system shall provide a web-based graphical user interface accessible through modern web browsers.
- The interface shall allow users to browse movies, select showtimes, choose seats (deluxe theaters only), and complete purchases with no more than 5 steps.
- The interface shall clearly display:
    - Movie title and description.
    - Theatre location.

    - Showtime.
    - Ticket prices (including applied discounts, if applied).
    - Seat availability with real time updates.
- The system shall provide a visible 5-min timer during checkout and ticket selection.
- The system shall provide clear confirmation messages for successful actions and clear error messages for issues.
- Error messages must be clear and descriptive, guiding users on how to fix the issues.
- The user interface shall support the following languages: English, Spanish, Swedish.
- The interface shall be usable on both desktop and mobile web browsers without requiring the download of an application.
- The user interface shall be simple to navigate.
- User interface shall not require the user to perform difficult tasks or actions, and should allow a first-time user to navigate the system without needing to be directed.

3.1.2 Hardware Interfaces

- The system shall operate on any device capable of running modern web browsers (ie. desktop computers, laptops, tablets, smart phones).
- Cinema staff shall use QR code scanners to validate digital tickets.
- The system shall support the printing of physical tickets at box office kiosks or at home to any device with a connected printer.

3.1.3 Software Interfaces

- The system shall interface with a centralized database which stores:
    - Movie listings
    - Showtimes
    - Seat availability
    - User accounts
    - Transaction history
    - System logs
    - Error logs
- The system shall work with external payment processing services which handle at least the following: Credit Card, PayPal, Bitcoin transactions.
- The system shall work with an email service provider to deliver digital tickets, receipts and confirmation messages.
- The system shall work with external review API services to retrieve movie ratings and critic quotes (IMDB, Rotten Tomatoes, etc.).

3.1.4 Communications Interfaces

- All communication between the client and server shall occur through HTTPS.
- The system shall support secure HTTP requests for user login, sign up, ticket purchasing, administrative requests, etc.
- The system shall send email confirmations through a secure email method.
- The system shall maintain session management through secure use of cookies.

3.2 Functional Requirements

This section describes specific features of the software project. If desired, some requirements may be specified in the use-case format and listed in the Use Cases Section.

3.2.1 <Concurrent User Support>

3.2.1.1 Introduction

This feature ensures that the system can handle at least 1000 users simultaneously without system failure or degradation of performance. This feature is important when it comes to high traffic hours, such as during the release of a popular movie or on weekend evenings.

3.2.1.2 Inputs

- Login Requests
- Ticket selection/purchase requests
- Movie browsing requests

3.2.1.3 Processing

- The system shall manage multiple connections concurrently to the server.
- The system shall maintain session integrity for all concurrent users.
- The system shall implement a queue mechanism for high-demand showings to regulate user access and prevent server overload.

3.2.1.4 Outputs

The system shall provide:

- Real-time seat availability for all users.
- Confirmation of successful login.
- Updated ticket availability reflected online.

3.2.1.5 Error Handling

The system shall:

- Display an error message if the maximum concurrent user threshold is exceeded.
- Prevent duplicate ticket purchases made by simultaneous requests.
- Relist incomplete/failed transactions immediately.
- Log all failures for administrator review.

3.2.2 <Web Browser Access>

3.2.2.1 Introduction

The system shall operate fully as a web browser application and shall not require installation of a mobile or desktop application.

3.2.2.2 Inputs

- HTTP(S) requests from user browsers
- User form submissions (login, purchases, feedback)
- Session cookies

3.2.2.3 Processing

The system shall:

- Render dynamic web pages through browser-compatible technologies.
- Ensure compatibility with modern browsers (Chrome, Safari, Edge, etc.)
- Maintain secure user sessions.

3.2.2.4 Outputs

- Interactive web interface.

- Secure transaction confirmation screens.

3.2.2.5 Error Handling
- Display error message if browser is not supported.
- Display connection error if server is unresponsive/unreachable.
- Log errors for administrator oversight.

3.2.3 <Bot Detection and Protection>

3.2.3.1 Introduction

The system shall detect, then protect from automated bot activity which is intended to mass purchase tickets to high-demand movies. Bots buy tickets in mass numbers and resell them, prevention protects real customers.

3.2.3.2 Inputs
- Login attempts.
- Purchase frequency data.
- User interaction timing data.
- CAPTCHA verification results.
- Attempts to purchase tickets in massive quantities.

3.2.3.3 Processing
- Require CAPTCHA verification at both login and checkout.
- Monitor abnormal requests and frequencies (ie. bulk purchases, excessive rapidity).
- Flag suspicious behavior patterns.
- Suspend accounts temporarily which exhibit bot-like behavior and request identity verification (ie. CAPTCHA) to regain access.

3.2.3.4 Outputs
- CAPTCHA trials.
- Account suspension notifications with steps to unsuspend.
- Alerts to administration.
- Log of bot activity.

3.2.3.5 Error Handling
- Deny transactions in case of failed CAPTCHA.
- Deny login requests in case of failed CAPTCHA.
- Log suspicious behavior.

3.2.4 <Database Interface Integration>

3.2.4.1 Introduction

The system shall interface with a centralized database that manages the movies, theaters, showtimes, ticket inventories, users, transactions, and administrator data.

3.2.4.2 Inputs
- Showtime lookup requests.
- Ticket purchase requests.
- Login requests (user and administrator side).
- Feedback submissions.

3.2.4.3 Processing

- Retrieve showtime and seat availability data from database.
- Retrieve ticket counts and pricing information.
- Immediately update ticket counts after purchase is successfully completed.
- Divide data (seats, showtimes, prices, etc.) by theater location.
- Store transaction history and user activity.
- The system shall maintain system-wide daily logs of all ticket purchases.

3.2.4.4 Outputs

- Updated seat availability.
- Updated ticket availability.
- Updated showtime information on website.
- Updated + accurate statistics of ticket sales for administrators.
- Transaction history and receipts available for users.

3.2.4.5 Error Handling

- Return transactions if database update fails, issue immediate refund.
- Display errors in case the database is unavailable.
- Log database failures.

3.2.5 <Ticket Purchase Constraints>

3.2.5.1 Introduction

The system shall enforce ticket quantity and time restrictions. The user can purchase no more than 20 tickets at a time, and the tickets shall only be available from two weeks prior to showtime and 10 minutes after the showtime begins.

3.2.5.2 Inputs

- Selected showtime.
- Number of tickets requested.
- Current system time and date.
- Payment information (credit card, PayPal, BitCoin, etc.)

3.2.5.3 Processing

- Validate that purchase does not exceed 20 tickets per transaction.
- Validate that showtime is within the two week booking window.
- Validate that purchase is not made later than 10 min. after showtime start.
- Start a checkout timer with (~5 min) upon ticket selection.
- Release reserved tickets upon timer expiration.
- Tickets may be returned or exchanged prior to showtime only if purchased through a registered user account.
- For deluxe theatres, the system shall allow users to select and reserve specific seat numbers.
- For regular theatres, the system shall track ticket quantity without assigned seating.

3.2.5.4 Outputs

- Purchase confirmations.
- Countdown timer display.
- Validation success or failure message.

3.2.5.5 Error Handling

- Display message if ticket limit exceeded.
- Display message if ticket booked outside valid window.

- Deny refund/exchange requests made after showtime begins.
- Cancel transaction when timer expires.
- Display message when timer expires.

3.2.6 <Administrator Mode Support>

3.2.6.1 Introduction

The system shall provide administrator controls and support for system management.

3.2.6.2 Inputs

- Administrator login requests.
- Administrator login credentials.
- Ticket override requests.
- User account management actions.
- Showtime update requests.
- Price update requests.
- Movie updates and overrides.

3.2.6.3 Processing

- Authentication of administrator credentials.
- Update movie listings, showtimes, prices, etc.
- Override of user purchase when necessary.
- Ability to block/unblock user accounts.
- Ability to temporarily suspend/unsuspend user accounts.

3.2.6.4 Outputs

- Updated system data.
- Sales reports for administration.
- Confirmation messages for successful admin actions.

3.2.6.5 Error Handling

- Deny access if authentication fails.
- Log unauthorized admin access attempts.
- Device and login verification tools to prevent unauthorized admin access.

3.2.7 <Customer Feedback System>

3.2.7.1 Introduction

The system shall collect and store customer feedback after ticket purchase.

3.2.7.2 Inputs

- User feedback rating (Smiley selection/Numerical 1-10).
- Optional text comments.

3.2.7.3 Processing

- Validate feedback submission.
- Store feedback in the database linked to the transaction and log.
- Rating aggregate values for administration report.

3.2.7.4 Outputs

- Confirmation of feedback submission.
- Thank you message post feedback submission.
- Updated feedback analysis and data for administrators.

3.2.7.5 Error Handling

- Display error message if feedback submission fails.
- Prevent duplicate feedback submissions for the same transaction.

3.2.8 <Discount Ticket Support>

3.2.8.1 Introduction
The system shall support discounted tickets and categories.
3.2.8.2 Inputs
- Selected discount type (ie. student, military, senior, etc.)
- Proof of eligibility verification.
- Promotional code validation.

3.2.8.3 Processing
- Validate purchase request.
- Validate discount eligibility.
- Apply fixed/percentage price reduction based on discount.
- Recalculate total cost.

3.2.8.4 Outputs
- Updated discount ticket price
- Confirmation of discount application.

3.2.8.5 Error Handling
- Display message if ineligible or eligibility unable to be verified.
- Prevent discount stacking unless permitted by admin.

3.2.9 <Online Review Scraping + Integration>

3.2.9.1 Introduction
The system shall retrieve and display critic ratings and quotes from external review sources (ie. Rotten Tomatoes, IMDB).
3.2.9.2 Inputs
- Movie title query
- External API/scraping request from review sources (ie. IMDB, Rotten Tomatoes) .

3.2.9.3 Processing
- Query external review platforms.
- Retrieve ratings and critic quotes.
- Refresh data every chosen interval (ie. 24 hrs).

3.2.9.4 Outputs
- Display ratings from reviews alongside movie listings.
- Display selected critic quotes.

3.2.9.5 Error Handling
- Display placeholder message if review retrieval fails.
- Log external API/scraping failures without affecting ticket sale mechanism.

3.2.10 <Multilingual Support>

3.2.10.1 Introduction
The system shall support the following languages: English, Spanish, and Swedish.
3.2.10.2 Inputs
- User language selection.

- Account language preference chosen upon sign-up.

3.2.10.3 Processing
- Load interface text based on selected language.
- Store language preference in account/session data.

3.2.10.4 Outputs
- Interface displayed in selected language (English, Spanish, Swedish)

3.2.10.5 Error Handling
- Default language set to English if language fails to load.
- Log language load failures.

3.2.11 <Loyalty Account Management>

3.2.11.1 Introduction

The system shall support user account registration and loyalty membership functionality.

3.2.11.2 Inputs
- User registration information.
- Payment information.
- Membership purchase request.
- Sign up/Sign in requests.
- CAPTCHA request.

3.2.11.3 Processing
- Store personal information securely.
- Store encrypted payment data.
- Track purchase history and tie to account details.
- Track account loyalty points.
- Allow membership card purchase.
- The system shall restrict concurrent logins to one active device per user account.

3.2.11.4 Outputs
- Account dashboard.
- Profile page.
- Loyalty point balance display.
- Membership confirmation statement.

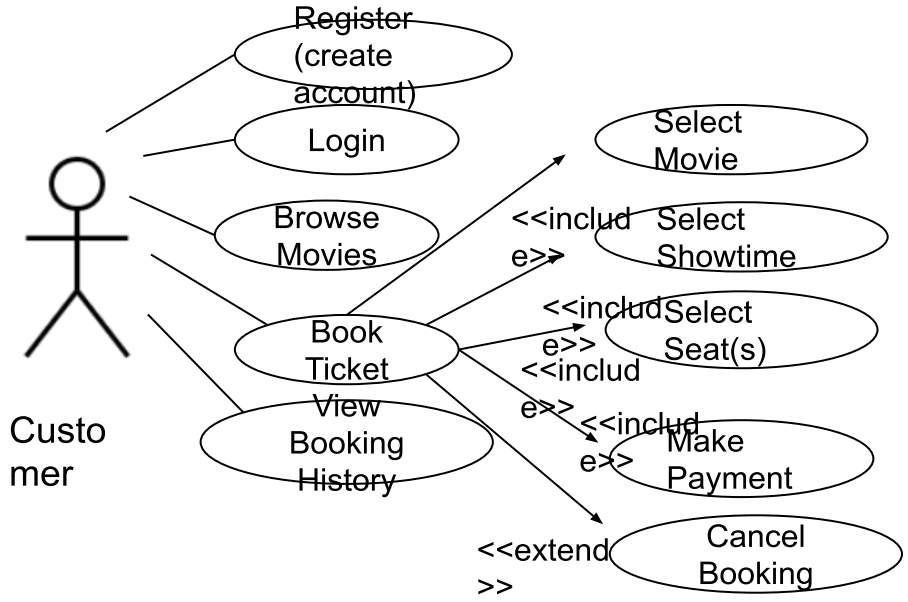
3.2.11.5 Error Handling
- Prevent duplicate accounts with identical credentials.
- Deny access if authentication fails.
- Log suspicious account activity.

3.3 Use Cases

3.3.1 Use Case #1 <Customer>:

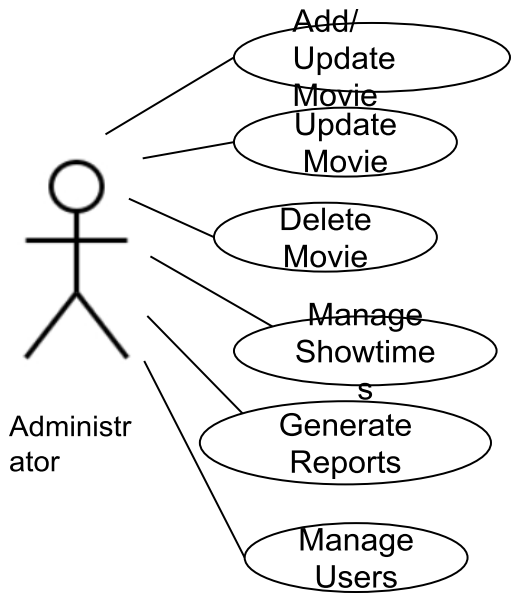
- Register/ Sign up: Create an account in order to access the system.
- Login: Access an existing account.
- Browse Movies: View the list of the current movies available.
- Book Ticket: Reserve seats for a selected movie, including the following steps:
    - Select Movie <<include>>: Choose a movie to watch.
    - Select Showtime <<include>>: Choose a showtime for the selected movie.

  - Select Seat(s) <<include>>: Choose preferred row and seat number/
  - Make Payment <<include>>: Complete the booking transaction through/
- View Booking History: List of the items in the cart/ the movie times selected.
- Cancel Booking <<extends>>: Cancel any purchased ticket, at least 1-2 hrs before showtime. This is optional and only occurs under certain conditions.



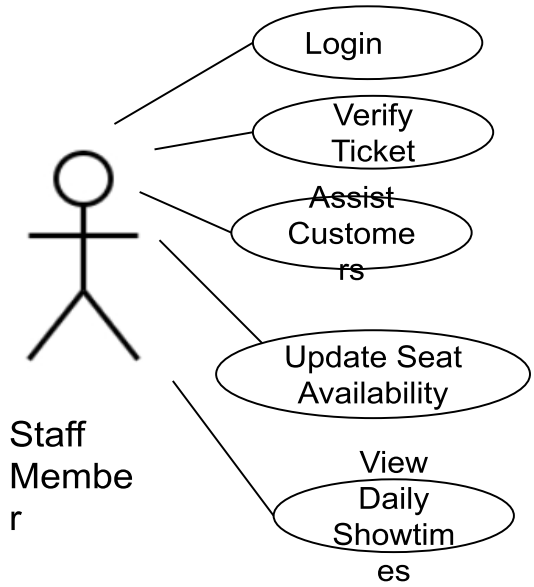
3.3.2 Use Case #2 <Administrator>:

- Add Movie: Add a new movie to the system, including the movie details (title, duration, genre, and rating)
- Update Movie: Edit movie details, such as scheduled times and description
- Delete Movie: Remove a movie from the system that is no longer available.
- Manage Showtimes: Schedule, update, or remove showtimes for movies.
- Generate Reports: Ticket sales, booking history, and user activity.
- Manage Users: Create, update, or remove staff/ administrator accounts as needed.



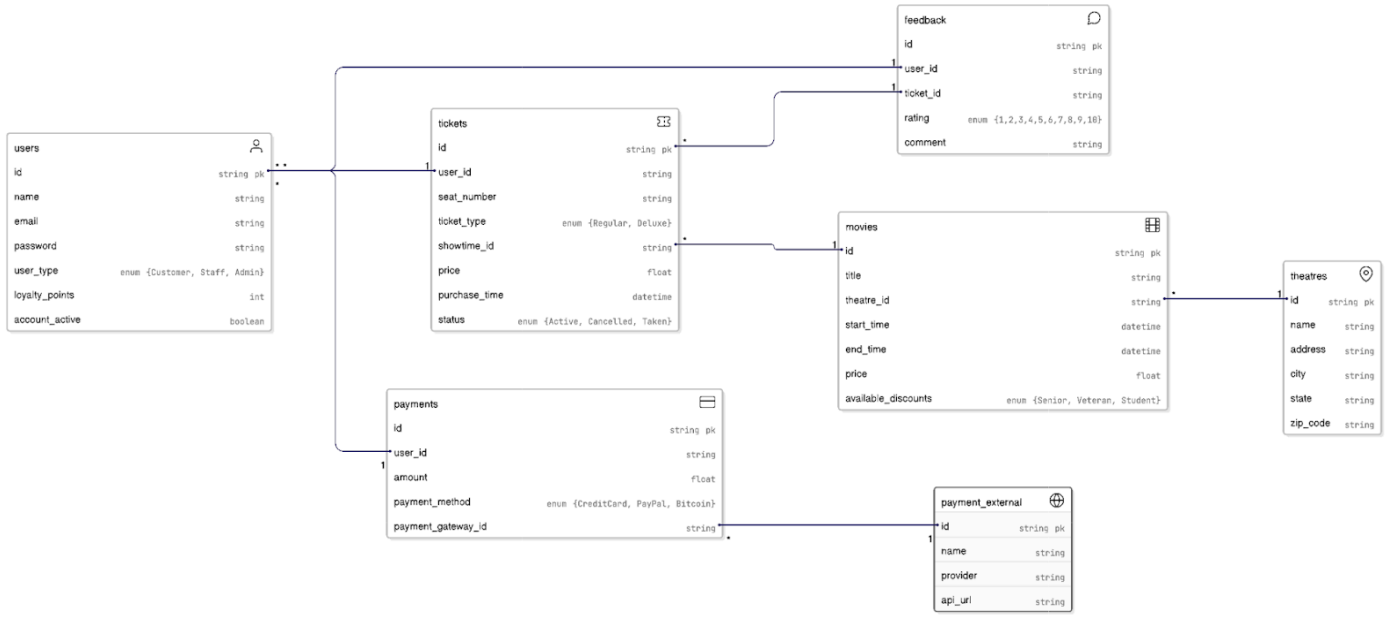
3.3.3 Use Case #3 <Staff Member>:

- Login: Login to access staff functions.
- Verify Ticket: Check confirmation code, in person, to validate booking.
- Assist Customers: Help customers with booking issues, cancellations, and general questions.
- Update Seat Availability: Adjust seat availability based on bookings and cancellations.
- View Daily Showtimes: View the scheduled movies for the day



3.4 Classes / Objects

UML Class Diagram:



3.4.1 <User>

3.4.1.1 Attributes
UserID → String
Name → String
Email → String
Password → String
User Type → Enum {Customer, Staff, Admin}
Loyalty Points → Int
Account Active → Boolean

3.4.1.2 Functions
- Login
- Logout
- Purchase tickets
- Submit feedback
- View tickets
- View loyalty points

3.4.2 <Ticket>

3.4.2.1 Attributes
Ticket ID → String
User ID → String
Showtime ID → String
Seat Number → String
Ticket Type → Enum {Regular, Deluxe}

Price → Float
Purchase Time → DateTime
Status → Enum {Active, Cancelled, Taken}

3.4.2.2 Functions
- Ticket purchase
- Ticket validation
- Ticket cancellation
- Ticket transfer

3.4.3 <Movie>

3.4.3.1 Attributes
Showtime ID → String
Movie Title → String
Theatre ID → String
Start time → DateTime
End Time → DateTime
Seat Map → Map <String, Boolean> (seat number, T/F isBooked)
Price → Float
Available discounts → Enum {Senior,Veteran,Student}

3.4.3.2 Functions
- Check availability of seat
- Reserve seat
- Book seat
- Apply discount rate

3.4.4 <Administrator>

Inherits from User class

3.4.4.1 Attributes
Showtime ID → String
Movie Title → String
Theatre ID → String
Start time → DateTime
End Time → DateTime
Seat Map → Map <String, Boolean> (seat number, T/F isBooked)
Price → Float
Available discounts → Enum {Senior,Veteran,Student}

3.4.4.2 Functions
- Update Movie
- Update Showtime
- Update Prices

- Override ticket
- Manage Users
- View analytics
- View logs

3.4.5 <Payment>

3.4.5.1 Attributes
Payment ID → String
User ID→ String
Amount → Float
Payment Method → Enum {CreditCard, PayPal, Bitcoin}

3.4.5.2 Functions
- Process payment
- Refund payment
- Link to external payment service

3.4.6 <Feedback>

3.4.6.1 Attributes
Feedback ID → String
User ID→ String
Ticket ID → String
Rating → Enum {1,2,3,4,5,6,7,8,9,10}
Comment → String

3.4.6.2 Functions
- Submit Feedback
- Get average rating

3.4.7 <Theatre>

3.4.7.1 Attributes
ID → String
Name → String
Address → String
City → String
State → String
Zipcode → String

3.4.7.2 Functions
- Update location

3.5 Non-Functional Requirements

Non-functional requirements may exist for the following attributes. Often these requirements must be achieved at a system-wide level rather than at a unit level. State the requirements in the

following sections in measurable terms (e.g., 95% of transactions shall be processed in less than a second, system downtime may not exceed 1 minute per day, > 30 day MTBF value, etc).

3.5.1 Performance

- The system shall support at least 1000 concurrent users without performance degradation.
- 95% of page loads shall complete within 2 seconds under normal operating conditions.
- Ticket purchase transactions shall complete within 5 seconds, excluding any third party processing delays.
- Seat availability shall update in real time in accordance with confirmed purchase (within 1 second).
- System shall implement a queuing mechanism for high-demand showings to prevent server overload.

3.5.2 Reliability

- The system shall maintain accurate seat inventory with 0 duplicate bookings.
- Confirmed ticket transactions shall not be lost in the event of a server failure.
- Daily logs of all ticket purchases shall be maintained for administrative review.
- Transaction failure rate must fall below 1%.

3.5.3 Availability

- The system shall maintain 99.99% time up during the theater operating hours.
- Scheduled maintenance shall not exceed 2 hours per month.
- In the event of a system outage, service restoration shall occur within 15 minutes of the outage.

3.5.4 Security

- All user data shall be transmitted using HTTPS encryption.
- Passwords shall be stored using hashed encryption methods.
- Each ticket shall contain a unique and non replicable identification number to prevent duplication or resale.
- The system shall restrict concurrent logins to one device per account.
- CAPTCHA verification shall be implemented to prevent bot automated purchase.
- Payment information shall be encrypted and not stored in plain text formatting.
- User sessions shall automatically expire after 15 minutes of inactivity.
- The user ticket checkout portal must automatically expire after 5 minutes.

3.5.5 Maintainability

- The system shall follow object oriented design principles, which will allow the individual components of the application (ie. ticketing, payment, feedback, emailing, etc.) to be updated independently.
- All the code shall be documented with inline comments and summarizations for maintenance and update.
- Error logs shall be detailed and provide valuable insight and information for administrators and developers to diagnose and resolve issues within a 24 hour period.
- System updates and maintenance shall be planned to minimize downtime, not exceeding 30 minutes/update cycle.

- The system shall allow administrators to adjust movie listings, showtimes, prices, etc. with ease and dynamic nature.

3.5.6 Portability

- The system shall operate entirely in a supported web browser, without requiring installation of an application.
- The system shall function correctly on the latest few versions of modern browsers (ie. Chrome, Safari, Edge, etc.)
- The system shall support both desktop and mobile browsers (cross-device functionality)
- The system shall be compatible with standard devices used by cinema staff for ticket validation (ie. QR code scanners, tablets, mobile phones, etc.)